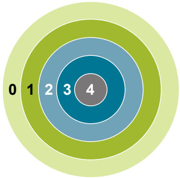
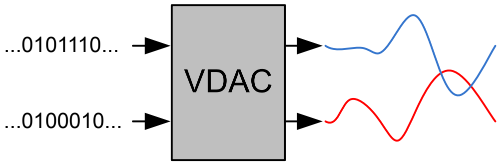
28. VDAC - Digital to Analog Converter





Quick Facts

What?

The VDAC is designed for low energy consumption, but can also provide very good performance. It can convert digital values to analog signals at up to 500 ksamples/second with 12-bit accuracy.

Why?

The VDAC can be used to generate accurate analog signals for sound, sensors and other applications, using only a limited amount of energy.

How?

The VDAC can generate high-resolution analog signals while the MCU is operating at low frequencies and with low total power consumption. Using the LDMA, the VDAC can be used to generate waveforms without any CPU intervention. The VDAC is available down to Energy Mode 3.

28.1 Introduction

The Voltage Digital to Analog Converter (VDAC) converts a digital value to an analog output voltage. It is useful in a number of different applications such as sensor excitation or low/medium frequency sound output. The VDAC has two rail-to-rail output channels that may act as independent DACs or be combined into a differential pair. The output buffers support high or low power operation as well as a high capacitance mode for driving loads up to 50 pF. Both channels have a 4-word FIFO for efficient throughput and minimum CPU intervention. Flexible conversion trigger sources allow for accurate AC and DC waveform timing, and a special sample-off mode can be used in conjunction with periodic output refresh to reduce energy consumption or act as a temporary excitation source.

28.2 Features

Each VDAC instance in a device supports the following features:

- Up to 500 ksps operation
- Two output channels
  - Can be combined into one differential output
- Integrated 7-bit clock prescaler with division factors ranging from 1 to 128
- Selectable voltage reference
  - Internal 2.5 V (effective full-scale)
  - Internal 1.25 V (effective full-scale)
  - AVDD supply
  - External VREFP pin
- Outputs available for internal and external use
  - Main low-impedance outputs to dedicated GPIO
  - Auxiliary outputs routable through ABUS to any ABUS-capable GPIO (higher impedance)
  - Internal routing of auxiliary outputs to other analog blocks
- Data conversion modes selectable per channel
  - Continuous Mode for high speed conversion or constant DC output
  - Sample-off Mode for per-sample conversion followed by channel disable
- Independent FIFO per channel
  - 4 word (12bit) depth for each channel
  - Programmable data valid level
  - Supports software flush
- Conversion trigger sources
  - Data write (software)
  - PRS input (synchronous and asynchronous)
  - Internal timer with power-of-2 selection from 2-64 prescaled clock cycles
- Refresh trigger sources
  - Refresh timer with power-of-2 selection from 2-256 low-frequency clock cycles
  - PRS input (synchronous or asynchronous)
- PRS Communication
  - Separate line for each channel
  - Sync and async PRS output pulse on finished conversion
  - PRS Level Output till channel is warmed
  - Async PRS Output Pulse on Refresh Timer Overflow and Internal Timer Overflow
- LDMA request on FIFO data valid level
  - Independent requests for each DAC channel
- Sine generation mode with differential support

28.3 Functional Description

An overview of the VDAC module is shown Figure 28.1 VDAC Block Overview Diagram on page 1079

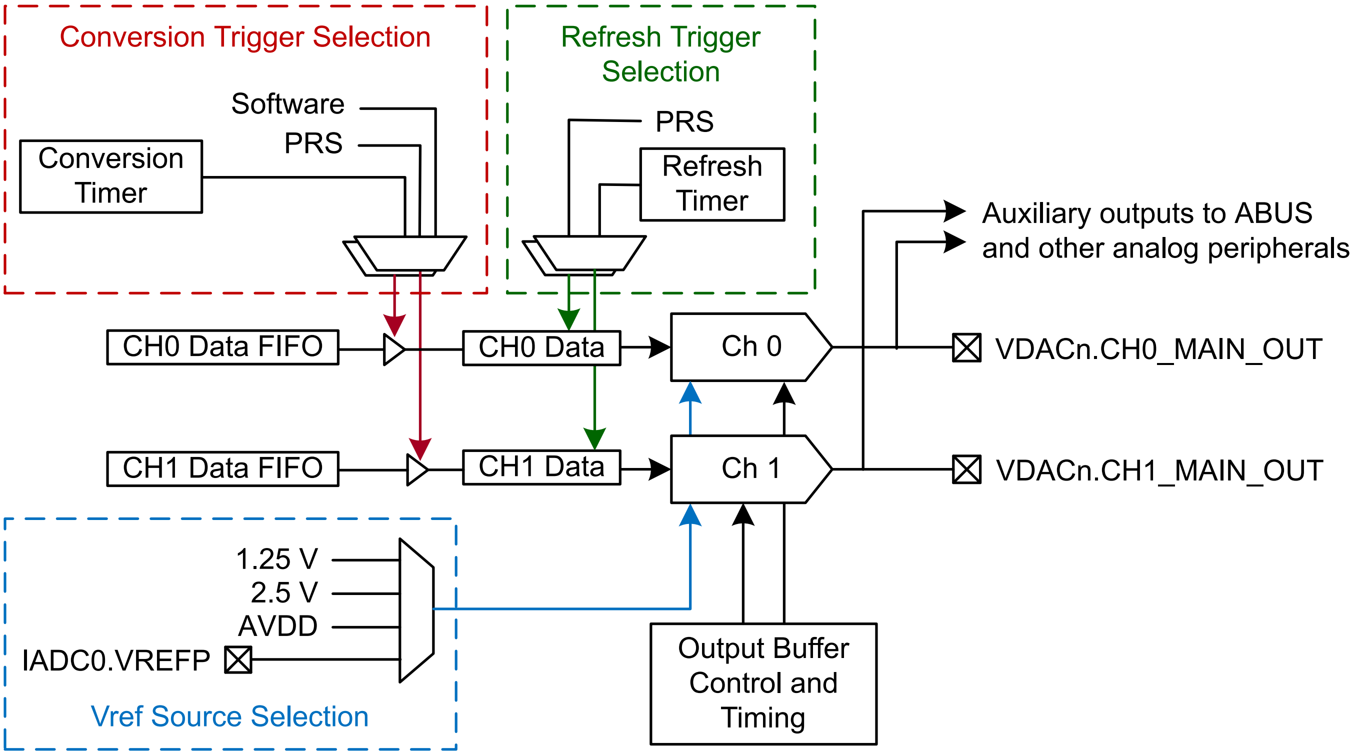


Figure 28.1. VDAC Block Overview Diagram

28.3.1 Power Supply

The VDAC module power (V_{VDAC}) is derived from the AVDD supply pin.

28.3.2 I/O Pin Considerations

The maximum usable analog signal that can be seen on external VDAC outputs depends on both the AVDD and IOVDD supplies. Specifically, the VDAC output will be limited to the lower of the two supply voltages on AVDD and IOVDD.

28.3.3 Enabling and Disabling a Channel

The VDAC module is enabled by writing 1 to EN in VDAC_EN.

A VDAC channel is enabled by writing 1 to the CHxEN and disabled by writing 1 to CHxDIS using in the CMD register. The channel status (enabled or disabled) can be read using the CHxENS bits in the STATUS register. The CHxENS bit will go high after a synchronization delay following a write to CHxEN. When disabling a channel the CHxENS bit will stay high until the VDAC channel is completely disabled.

Software should configure the VDAC before enabling a channel. The following registers are used to configure all the available features of the VDAC:

- VDAC_CFG
- VDAC_CH0CFG
- VDAC_CH1CFG
- VDAC_OUTTIMERCFG

A VDAC channel will not begin driving its output before it is enabled *and* has received a conversion trigger or refresh trigger. After a channel is enabled, it will listen for either conversion trigger sources specified in TRIGMODE in VDAC_CHxCFG or refresh trigger sources specified in REFRESHSOURCE in VDAC_CHxCFG. If TRIGMODE is set to SW and the CHxF FIFO is not empty, a conversion will start immediately when the channel is enabled. When disabling a channel, any pending triggers are flushed.

When disabling the VDAC module, user code must poll the status bit VDAC_EN.DISABLING to ensure that the module is cleanly reset and back to its initial condition.

28.3.4 Clock Selection

The VDAC logic accepts three clock sources from the CMU: LSPCLK, VDACn_CLK, and VDACn_REFRESH_CLK. The APB register interface and FIFO write logic are clocked from the LSPCLK. The rest of the VDAC state machine is clocked mainly by a prescaled version of VDACn_CLK. VDACn_DAC from the CMU can be up to 80 MHz. The PRESC bit field in the CFG register should be set to scale VDACn_CLK to no more than 1 MHz. The VDACn_REFRESH_CLK is a low-frequency clock source which only clocks the dedicated refresh timer.

The clock request for VDACn_CLK to the CMU from VDAC is on-demand by default. This means the VDAC core clock is gated off most of the time except:

- New Conversion or Refresh Trigger
- Sine Generation Active Window
- VDAC_CHxCFG.TRIGMODE = SYNCPRS
- VDAC_CHxCFG.REFRESHSOURCE = SYNCPRS
- VDAC_CHxCFG.TRIGMODE = SW and the VDAC CHx FIFO is not empty
- VDAC_CHxCFG.TRIGMODE = INTTIMER

In some cases it is necessary or preferred to have the VDAC clock active all the time. To turn off on-demand clocking, the VDAC_CFG.ONDEMANDCLK bit can be set to '1', which always requests VDACn_CLK from the CMU. On-demand clocking should be disabled if EM23GRPACLK is selected for VDACn_CLK.

If the VDAC will only perform conversions in EM0 and EM1, any clock source for the VDAC may be used.

If the VDAC is to be operated in EM2 or EM3, VDACn_CLK must be configured to use either HFRCOEM23, EM23GRPACLK or FSRCO instead of the EM01GRPACLK clock. FSRCO is the fast start oscillator which starts quickly but is not as accurate as the other oscillators. HFRCOEM23 is generally recommended for EM2/EM3 operation. When using FSRCO or HFRCOEM23, the clock source can be selected to be "on demand" so as not to waste current when the DAC is not doing a conversion. On demand clocking is configured by setting VDAC_CFG.ONDEMANDCLK to 0.

EM23GRPACLK is recommended only when VDAC is expected to do very slow sample conversions/refresh. VDAC_CFG.ONDEMANDCLK should be set to 1 if EM23GRPACLK is selected as the VDAC clock source.

Note: When HFRCOEM23 is selected as a clock source to VDAC and clocking is on-demand (CFG.ONDEMANDCLK = 0), HFRCOEM23 on-demand clocking must be enabled. This allows the power domain which powers HFRCOEM23 to be ON during EM2 and the clock request to HFRCOEM23 can be honored.

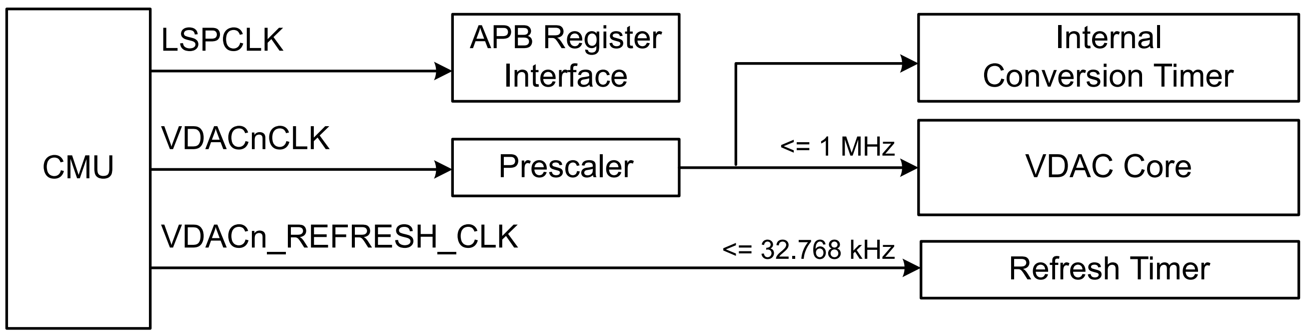


Figure 28.2. VDAC Block Clocks Diagram

28.3.4.1 Internal Clock Prescaler

The VDAC has an internal clock prescaler, which can divide the VDACn_CLK input clock by any factor between 1 and 128, by setting the PRESC field in the CFG register. The resulting prescaled clock is 50% duty cycle and is used by the converter core. The frequency is given by Figure 28.3 VDAC Clock Prescaling on page 1082 :

$$f_{CLK\_DAC\_PRESC} = f_{VDACn\_CLK} / (PRESC + 1)$$

Figure 28.3. VDAC Clock Prescaling

where f_{VDACn_CLK} is the input clock frequency coming from the CMU. The $f_{CLK\_DAC\_PRESC}$ must be programmed to be at most 1 MHz.

The prescaler runs continuously when the VDAC is enabled or when VDAC_SWRST.SWRST is set. When running with a prescaler setting higher than 0, there can be an unpredictable delay from the time a conversion was triggered to the time the actual conversion takes place. This is because the conversions are controlled by the prescaled time base, and the conversion can arrive at any time during a prescaled clock (CLK_DAC_PRESC) period. A second reason for unpredictable delay between a trigger and the associated conversion is that the activity on one channel can impact whether the VDAC reference is warm or not - when two channels are used independently with warmup, it can impact whether a warmup is required on a trigger.

The uncertainty related to the clock prescaler can be addressed by using CH0PRESCRST. If the CH0PRESCRST bit in VDAC_CFG is set, the prescaler will be reset every time a conversion is triggered on channel 0. This leads to a predictable latency between channel 0 triggers and conversions (assuming the warmup sequence is deterministic as well). If channel x is used in continuous mode, the warm-up sequence will only apply once when the channel is enabled and software can use the VDAC_STATUS.CHxWARM bit to determine if the VDAC has warmed up.

28.3.4.2 VDACn_REFRESH_CLK

VDAC also has another clock coming in from the CMU, VDACn_REFRESH_CLK. This clock is asynchronous to the VDACn_CLK and is used to clock the internal refresh timer. VDACn_REFRESH_CLK is connected directly to the EM23GRPACLK source in the CMU and hence it is a low-frequency clock sourse with a maximum frequency of ~32 kHz. The refresh timer inside VDAC is a slow running, low power timer. When this refresh timer is used as a refresh trigger source and the timer overflows, it will trigger a VDACn_CLK on-demand request from the CMU.

28.3.5 Conversions

The VDAC consists of two channels (channel 0 and 1) with separate 4 deep FIFOs with 12-bits data elements (VDAC_CHxF.DATA). These can be used to produce two independent single ended outputs or the channel 0 register can be used to drive both outputs in a differential mode. The VDAC supports two conversion modes: **continuous** and **sample-off.**

Continuous Mode

In continuous mode the VDAC buffers will remain on, and channels will drive their outputs continuously till the channel is disabled with the data in the VDAC_CHxF.DATA register. A channel is configured in continuous mode by programming the CONVMODE bitfield in VDAC_CHxCFG to CONTINUOUS. This mode will maintain the output voltage from a conversion indefinitely, until a new output is triggered or until the channel is disabled.

In continuous mode the CHxOUTHOLDTIME field in VDAC_OUTTIMERCFG should be programmed to zero to achieve the maximum update rate. Both these settings need to be configured before VDAC module is enabled.

Sample-off Mode

In sample-off mode the VDAC will only drive the output for a limited time per conversion. A channel is configured in sample-off mode by programming the CONVMODE bitfield in VDAC_CHxCFG to SAMPLEOFF. The CHxOUTHOLDTIME field in the VDAC_OUTTIMERCFG register determines how long the output will be driven after a conversion or refresh trigger occurs. The VDAC will drive the output for CHxOUTHOLDTIME number of CLK_DAC_PRESC cycles before tri-stating the output again (and therefore if CHxOUTHOLDTIME is set to zero, the output will never be driven when using sample-off mode). Both these settings need to be configured before VDAC module is enabled.

28.3.6 Conversion Trigger

Conversions can only be performed while a channel is enabled and the CHxF FIFO is not empty, see 28.3.3 Enabling and Disabling a Channel.

- If CHxCFG.TRIGMODE is programmed to SW, a conversion can be started automatically when the CHxF.DATA is not empty. Writing to the FIFO will trigger a conversion on the specified channel.
- If CHxCFG.TRIGMODE is programmed to SYNCPRS or ASYNCPRS, a conversion can be started by an incoming pulse on the selected PRS channel. VDAC expects a PRS pulse coming from both synchronous and asynchronous PRS producers. Depending on the TRIGMODE and channel enable, the VDAC will process the PRS pulse from the respective producer.
- If CHxCFG.TRIGMODE is programmed to INTERNALTIMER, the internal timer is uses to start conversions, and a new conversion will start on any overflow of the internal timer. See 28.3.13 Internal Timers.

28.3.7 Refresh Trigger

The refresh mechanism can be used to periodically refresh the output with the most recent conversion result. A refresh trigger can only happen while a channel is enabled, see 28.3.3 Enabling and Disabling a Channel.

- If CHxCFG.REFRESHSOURCE is programmed to SYNCPRS or ASYNCPRS, a refresh can be started by an incoming pulse on the selected PRS channel. The VDAC refresh mechanism requires a PRS pulse for either configuration of CHxCFG.REFRESHSOURCE. Depending on the REFRESHSOURCE and channel enable, VDAC processes the PRS pulse from the respective producer.
- If CHxCFG.REFRESHSOURCE is programmed to REFRESHTIMER, a conversion will start on an overflow of the refresh timer. See 28.3.13 Internal Timers.

Note: The refresh mechanism never pops a new conversion from the FIFO. It just re-triggers the conversion based on the last data converted by the conversion trigger to generate the same voltage output at the load. A periodic refresh of the output voltage into a suitable RC filter (to reduce ripple) may be a way to establish a low-energy DC bias in some applications.

Conversion Trigger and Refresh Trigger co-existence

- Conversion triggers are given preference over refresh triggers. During a conversion trigger sample conversion, if a refresh trigger occurs it is ignored.
- Refresh triggers are served only when the conversion trigger sample conversion is completely finished.
- During a refresh trigger sample conversion, if a conversion trigger occurs, it is held by the VDAC and served once the refresh trigger conversion is finished. The VDAC never ignores conversion triggers.

28.3.8 PRS Communication

PRS triggers can be used to set a constant sample frequency, for instance by using a TIMER, or to synchronize conversion events with other hardware. In order to get a jitter-free sample rate from a PRS source, set CHxCFG.TRIGMODE to SYNCPRS and set the CFG.CH0PRESCRST bit to ensure the prescaler is reset on trigger. Note that because the prescaler is shared between channels, this is only possible for channel 0.

For CHxCFG.TRIGMODE / CHxCFG.REFRESHSOURCE of ASYNCPRS, the sample frequency cannot be guaranteed to be jitter-free with respect to the PRS pulses. The CH0PRESCRST bit in VDAC_CFG can still be set to reset the clock prescaler on every PRS trigger for better predictability. Note, this can be set only on channel 0.

The Input PRS frequency should never be higher than 0.5 MHz (the fastest possible sample rate). In addition, the input PRS frequency should not be higher than f_{VDACn_CLK} / 4 (in synchronous mode). If the PRS frequency is set too high, some PRS pulses may be dropped and the output can jitter.

VDAC also sends out the following list of output signals as a PRS producer:

- CHxDONEASYNC - CHx Conversion Done Asynchronous PRS Pulse Output
- CHxWARM - CHx Output Valid Async PRS Level Output
- CHxDONESYNC - CHx Conversion Done Sync PRS Pulse Output
- REFRESHTIMEROF - Refresh Timer Overflow Async PRS Pulse Output
- INTERNALTIMEROF - Internal Timer Overflow Async PRS Pulse Output

28.3.9 Reference Selection

The VDAC supports four voltage reference options:

- Internal 1.25 V Reference (effective full scale)
- Internal 2.5 V Reference (effective full scale)
- AVDD
- External IADC0.VREFP Pin

The voltage reference is selected by programming the REFRSEL field in VDAC_CFG, and is shared for both VDAC channels. The selected voltage reference sets the full scale output voltage of the VDAC. In the case of the internal 1.25 V and 2.5 V reference options, the VDAC uses a lower internal voltage as the reference, and scaling techniques to produce an effective full scale voltage of 1.25 V or 2.5 V respectively. The 2.5 V internal reference can still be used over the entire supply voltage range of the device without any dropout. However, the output will be limited by the supply rail(s) and VDAC will not be able to drive above the supply.

28.3.10 Power Modes

The VDAC supports independently-selectable high power and low power modes per channel. Each has a maximum load capacitance of 50 pF. The power mode for each channel is configured by setting VDAC_CHxCFG.POWERMODE. By default, the VDAC starts in HIGHPOWER mode.

VDAC also supports an additional high capacitance load mode in conjunction with the HIGHPOWER power mode. High capacitance mode is enabled by setting VDAC_CHxCFG.HIGHCAPLOADEN to 1. The minimum load capacitance on the pin is 25 nF for this mode.

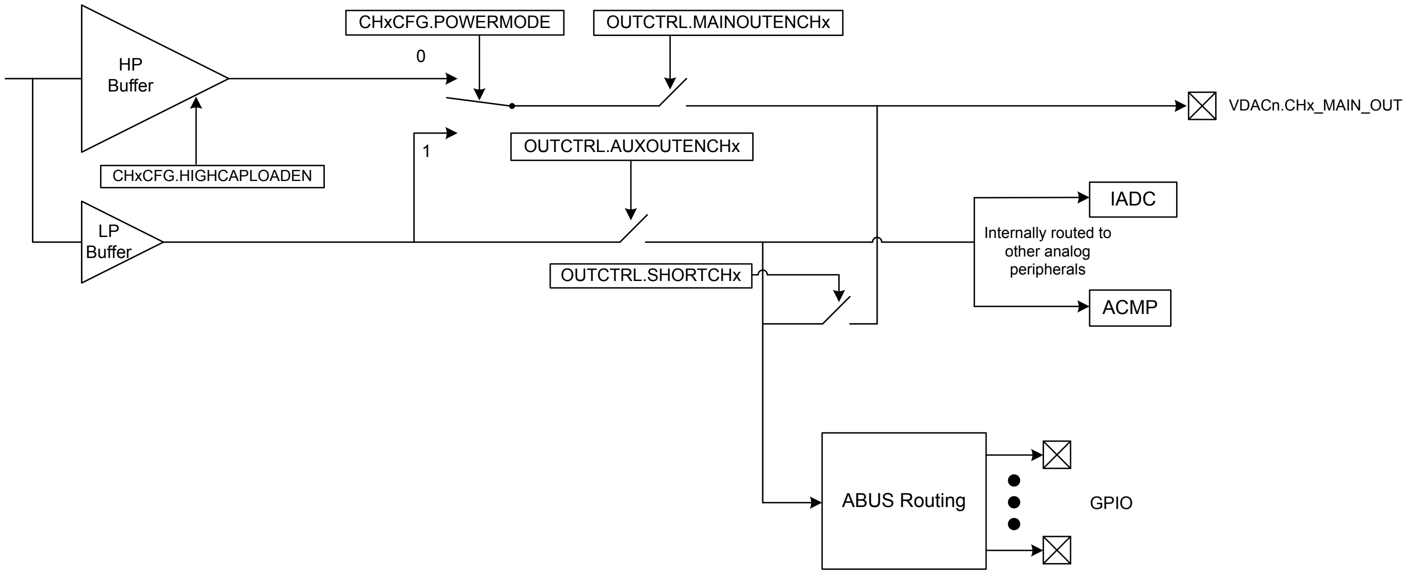


Figure 28.4. VDAC Block Power Modes Diagram

28.3.11 Warmup Time and Initial Conversion

When a channel is enabled, it needs to warm up. This is performed automatically when the channel is enabled if the CONVMODE in VDAC_CHxCFG is set to Continuous mode.

If the CONVMODE in VDAC_CHxCFG is set to Sample-off mode, then every sample conversion is preceded by warmup of the VDAC. Hence, the sample conversion time in case of sample-off mode includes the WARMUPTIME of the VDAC too.

The time allocated for warmup depends on the WARMUPTIME field in VDAC_CFG. WARMUPTIME is specified in prescaled VDACn_CLK cycles and should be set to a minimum of 3 us (3 clocks if CLK_DAC_PRESC = 1 MHz). Software is responsible for programming the correct value to WARMUPTIME before enabling a channel. If the time is programmed too short, an undefined voltage may be output until the analog portion of VDAC settles.

The CHxWARM bits in VDAC_STATUS are set when the warmup period has completed. A consequence of the warmup period is that in continuous mode, the recommended programming model is to either wait for CHxWARM status before starting conversions in order to make sure all samples have the same timing, or perform a dummy conversion to make the VDAC settle to a known voltage first.

28.3.12 Output Mode

The two VDAC channels can act as two separate single-ended channels or be combined into one differential channel. This is selected through the DIFF bit in VDAC_CFG.

- **Single Ended Output**

When operating in single ended mode, the channel 0 output is on VDAC_OUT0 and the channel 1 output is on VDAC_OUT1. The output voltage can be calculated using Figure 28.5 VDAC Single Ended Output Voltage on page 1085

$$V_{OUT} = V_{VDACn\_OUTx} - V_{SS} = V_{ref} \text{ x CHxDATA/4096}$$

Figure 28.5. VDAC Single Ended Output Voltage

where CHxDATA is a 12-bit unsigned integer.

- **Differential Output**

When operating in differential mode, both VDAC outputs are used to produce a differential output. The conversion uses the VDAC_CH0DATA buffer as the data source. The data written to VDAC_CH0DATA is interpreted as a two's complement number, with the MSB of the 12-bit value being the sign bit. Conceptually, a 'positive' and 'negative' output, with a common-mode voltage of $V_{ref}/2$ are generated from this information.

The positive output appears on VDAC_CH1 and the negative output appears on VDAC_CH0. The output voltage can be calculated using Figure 28.6 VDAC Differential Output Voltage on page 1085:

$$V_{OUT} = V_{VDACn\_CH1} - V_{VDACn\_CH0} = V_{ref} \text{ x CH0DATA/2048}$$

Figure 28.6. VDAC Differential Output Voltage

where CH0DATA is a 12-bit signed integer. The common mode voltage is $V_{ref}/2$.

When using differential mode, the user needs to program only CH0 settings to reflect to both channels. Any programmed settings in CH1 will be ignored in differential mode.

28.3.13 Internal Timers

VDAC has two internal timers that run independently of each other:

Conversion Timer

VDAC includes an internal conversion timer.This conversion timer is automatically started if a channel selects INTERNALTIMER as TRIGMODE in VDAC_CHxCFG reigster and the channel is enabled. The conversion timer will count the number of prescaled VDACn_CLK cycles programmed in VDAC_CFG.TIMEROVERFLOWPERIOD before wrapping and generating a conversion trigger. The timer overflow period is from 2-64 prescaled VDACn_CLK cycles. The conversion timer overflow automatically initiates the sample conversion if the CHxF FIFO is not empty.

INTERNALTIMEROF is also available as an asynchronous PRS signal, generating a pulse that lasts for 1 prescaled VDACn_CLK cycle. Note that since this timer runs off prescaled VDACn_CLK, the VDACn_CLK is always requested from the CMU when the internal conversion timer is used.

Refresh Timer

VDAC includes an internal low power refresh timer.This refresh timer is automatically started if a channel selects REFRESHTIMER as REFRESHSOURCE in VDAC_CHxCFG register and the channel is enabled. The refresh timer will count the number of VDACn_REFRESH_CLK cycles programmed in VDC_CFG.REFRESHPERIOD before wrapping and generating a refresh trigger. The refresh period is from 2-256 VDACn_REFRESH_CLK cycles. The refresh timer overflow automatically initiates the refresh.

REFRESHTIMEROF is also available as an asynchronous PRS signal, generating a pulse that lasts for 1 VDACn_REFRESH_CLK cycle. Note that since this timer runs off VDACn_REFRESH_CLK, which is asynchronous to VDACn_CLK, VDACn_CLK is only requested from the CMU when the refresh timer overflows. Also, as discussed in 28.3.4 Clock Selection, this timer always runs off a slow clock, hence this refresh conversion is low power.

28.3.14 FIFO

VDAC has two FIFOs, one for each channel. Each FIFO is 4 samples deep. Data is pushed into the FIFO by either the CPU or LDMA, and happens on the bus clock. Samples are popped from the FIFO on the VDACn_CLK domain by the VDAC core whenever there is a new conversion trigger. Both FIFOs support a data flush from the CPU.

CHxF FIFO Programming Model

Before enabling VDAC, set the low threshold Data Valid Level for each channel by programming VDAC_CHxCFG.FIFODVL Bit. If in DIFF Mode, the channel 1 DVL settings are ignored. Fill the VDAC_CHxF.DATA with the number of entries greater than programmed DVL to avoid triggering of DVL Interrupt. Details of the DVL Interrupt are explained in 28.3.18 Interrupts and Wakeup. The number of valid entries per channel can be read from VDAC_STATUS.CHxFIFOCNT (only if VDAC_CFG.ONDEMANDCLK is set to 1). The status of whether the FIFO is full or empty can be read from VDAC_STATUS.CHxFIFOFULL and VDAC_STATUS.CHxFIFOEMPTY respectively.

CHxF FIFO Flush Programming Model

In case there is incorrect data programmed in the FIFO or in the event of a FIFO overflow, the CPU can issue a flush by programming VDAC_CMD.CHxFIFOFLUSH. This flush bit resets both the write pointer (in the bus clock domain) and the read pointer (in rhe CLK_DAC domain) together. During a flush, new conversions should not be triggered on the channel whose FIFO contents are flushed. The flush status is reported with the VDAC_STATUS.CHxFIFOFLBUSY bit, which will remain high during the flush operation and return low when it is complete. Since the CPU issues flush to the CHx FIFO, there are several steps that should be followed in order to avoid any spurious voltage at VDAC Output:

- Disable the channel using VDAC_CMD.CHxDIS = 1
- Disable the VDAC channel in the LDMA to avoid any new CHxFIFO writes during flushing
- Issue the flush command and poll the VDAC_STATUS.CHxFIFOFLBUSY status bit
- After flushing is complete, re-enable the VDAC channel and optionally re-program the LDMA

28.3.15 Keepwarm Sub-modes

VDAC has two keepwarm modes that can be used with Sample-off Conversion Mode to define the behaviour of the analog portion of the VDAC between sample conversions. The keepwarm mode needs to be set along with Sample-off Conversion Mode before enabling the VDAC module. The two options are:

Bias Keepwarm

Bias Keepwarm Mode is primarily used to reduce kickback to the reference bias, in case it is shared with IADC. This mode can be activated by setting the VDAC_CFG.BIASKEEPWARM Bit to 1. During this mode, the VDAC Analog Bias remains ON in between sample conversions instead of shutting down after the hold time expires.

This mode is only relevant when using the internal 1.25 V or 2.5 V reference selections. This mode does not reduce the required warm-up time of the VDAC. Enabling this mode will typically cost about 4 uA of additional current when the VDAC is idle. The VDAC analog bias remains on irrespective of the channel enable/disable, until the VDAC module is disabled.

Channel Keepwarm

Channel Keepwarm Mode is primarily used to reduce kickback between the two channels, or to reduce the start-up time of the VDAC in-between sample conversions. This mode is activated per channel by setting the VDAC_CHxCFG.KEEPWARM bit to 1. During this mode, the VDAC remains warmed up in-between sample conversions and can convert new samples without incurring a warm up delay. This mode can only be used with Sample-off conversion mode. Setting this mode typically costs about 10-20 uA of additional current when the VDAC is idle.

28.3.16 LDMA Interface

The VDAC has two FIFOs (one per channel) which can be filled using the LDMA whenever there is space available in the FIFO. To facilitate this, the VDAC generates a DMA request per channel to the LDMA when the FIFO count reaches the lower threshold Data Valid Level (DVL). DVL is set by programming VDAC_CHxCFG.FIFODVL to set the watermark.

DMA REQ and SREQ

Both Channel 0 and 1 generate DMA requests in the bus clock domain only when the VDAC_CMD.CHxEN bit is set. In the case of DIFF mode, Channel 1 will not generate LDMA requests. The FIFOs are initially empty, hence as soon as the channels are enabled, each channel will send out a DMA REQ. The request is cleared when FIFO is filled beyond DVL+1.

If there is at least 1 valid space in the FIFO and the DMA channel is not active, each VDAC channel will also send a SREQ to LDMA. This SREQ is also gated with the channel enable.

EM2 Operation

When the system is in EM2, the VDAC can generate new DMA requests as well. This feature can be enabled by setting the VDAC_CFG.DMAWU bit to 1 before enabling VDAC. When enabled and the FIFO count in the read domain falls below the programmed DVL setting, the VDAC will generate a request to the EMU to enter an EM1 state. Once the EMU enters this state and the bus clock is available, VDAC generates the DMA REQ/SREQ per channel to request new data for conversion.

The VDAC keeps the request for data high until the FIFO count is above the programmed DVL setting. Once the condition is met, VDAC automatically pulls the request low and the system can return fully to EM2.

28.3.17 Sine Generation Mode

The VDAC contains an automatic sine-generation mode, which is enabled by setting the SINEMODE bit in VDAC_CFG. In this mode, the VDAC data from the FIFO is overridden with data from an internal hard-coded sine lookup table.

Sine mode is supported only for the fastest configuration of the VDAC in continuous mode. Therefore the CONVMODE bit in VDAC_CH0CFG needs to be set to CONTINUOUS and the CH0OUTHOLDTIME bit in VDAC_OUTTIMERCFG needs to be programmed to zero for channel 0 to use sine generation mode. The TRIGMODE and REFRESHSOURCE bitfields in VDAC_CHxCFG need to be programmed to NONE for channel 0 in order to avoid interference in sine output generation from other triggers. Other trigger modes are not supported. The SINE wave will always be output on channel 0 and therefore requires that this channel is enabled by writing 1 to CH0EN in the VDAC_CMD register. If DIFF is set in VDAC_CFG, the sine wave will additionally be output on channel 1, but inverted. Note that the first sample will only be available after the CHxWARMED bit is 1 in VDAC_STATUS register after setting CH0EN =1 in VDAC_CMD.

Each period, starting at 0 degrees, is made up of 16 samples and the frequency is given by Figure 28.7 VDAC Sine Generation Frequency on page 1087.

$$f_{sine} = f_{CLK\_DAC\_PRESC} / 32$$

Figure 28.7. VDAC Sine Generation Frequency

When DIFF is 0 and SINEMODE is 1 in VDAC_CFG, a sine wave will be generated on channel 0 but channel 1 can still be used independently as a single-ended DAC output. Channel 1 settings are ignored if DIFF is 1.

To configure the VDAC for sine wave output:

- Set VDAC_CFG.SINEMODE to 1 to put the VDAC in sine wave mode
- Set VDAC_CH0CFG.CONVMODE to CONTINUOUS
- Set VDAC_CH0CFG.TRIGMODE to NONE
- Set VDAC_CH0CFG.REFRESHSOURCE to NONE
- Set VDAC_OUTTIMERCFG.CH0OUTHOLDTIME to 0
- Configure VDAC_CFG.DIFF for single-ended or differential output
  - 0 = Single ended output (only channel 0 is used)
  - 1 = Enable differential output (both channel 0 and channel 1 will be used)
- Configure VDAC_CFG.SINERESET for desired behavior when halted
  - 0 = Sine wave output is not reset when halted, and will continue at the next sample when re-started
  - 1 = Sine wave output is reset to 0 degrees when halted, and output will return to Vref / 2
- Set VDAC_CFG.CH0EN to 1 to enable the VDAC module

28.3.17.1 Sine Generation - Software Control

The sine signal generation may be controlled by software using the SINEMODESTART and SINEMODESTOP commands in VDAC_CMD. When SINEMODESTART is set to 1, sine wave generation will be started. Setting SINEMODESTOP to 1 will halt the sine wave generation. If SINERESET in VDAC_CFG is set to 1, the sine output will be reset to 0 degrees when the SINEMODESTOP bit is set to 1, resulting in a voltage of Vref / 2 on the output channel(s). If SINERESET equals 0, a SINEMODESTOP command will stop progress of the sine wave at the sample currently being output. The sine will continue at the next sample when SINEMODESTART is issued again.

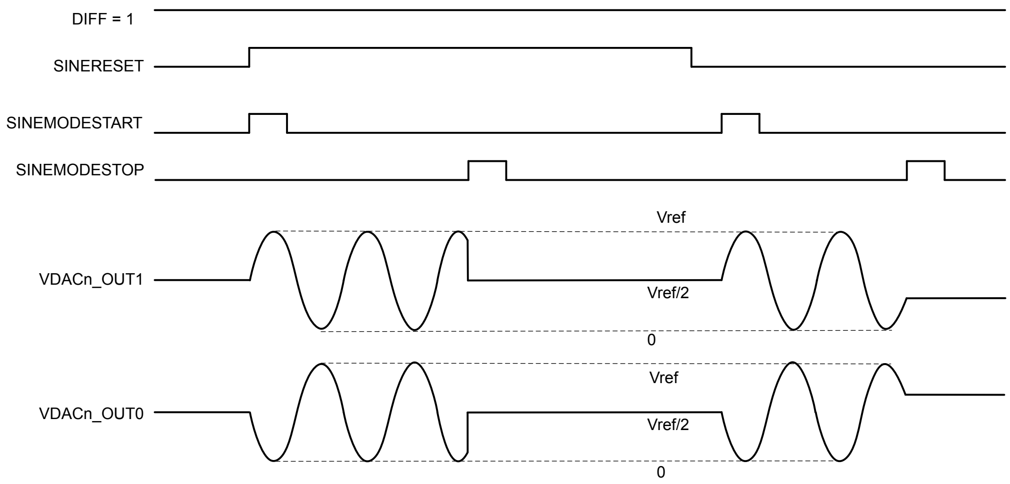


Figure 28.8. VDAC Sine Mode - Software Control

28.3.18 Interrupts and Wakeup

The VDAC has several interrupt flags in the VDAC_IF register, indicating various state and error conditions related to output, FIFO status, and conversions. Each channel has a separate set of interrupt flags.

- **Conversion Done**

  The Conversion Done (CHxCD) interrupt flags are set when a conversion is complete. The flags are set after a channel has driven the output with the programmed code.
- **Data Valid Level (DVL)**

  The Data Valid Level (CHxDVL) interrupt flags are set when the FIFO Count reaches below or equal to Data Valid Level (DVL). The DVL number is programmed in CHxCFG.DVL Register Bits. These flags are initially set, and gets cleared when CHxF FIFO is filled with new set of data, thereby making count >= DVL. The FIFO Count can be read through VDAC_STATUS.CHxFIFOCNT Flag. Note that this fifo count is generated in the read domain, hence expect read synchronization delay for the count value.
- **Overflow/Underflow**

  If a write is attempted in CHxF FIFO whilst it is full, the channel overflow flag (CHxOF) will be set. If a new conversion is triggered (e.g. via PRS) before whilst CHxF FIFO is empty, the channel underflow flag (CHxUF) will be set. The FIFO full and empty status can be read through VDAC_STATUS.CHxFIFOFULL and VDAC_STATUS.CHxFIFOEMPTY Flags.
- **ABUS Conflict/Allocation Error**

  In case both channel 0 and channel 1 request the same Port and Pin for conversion output through ABUS, an ABUS Conflict Error IF (ABUSINPUTCONFLICT) will be set. In case the ABUS Allocation is not coherent with Programmed Port and Pin Setting for either channel, ABUS Allocation Error interrupt flag (ABUSALLOCERR) will be set.

Not all interrupt flags will wake up EMU to EM0 when the VDAC is operating in EM2. Only those interrupts that need interaction with the host will wakeup the CPU. For this, DVL, ABUSALLOCERR and/or ABUSINPUTCONFLICT are used as wakeup interrupt source too.

28.3.19 VDAC Output Configuration

The VDAC analog outputs can be routed either to specific fixed pins, to configurable GPIO through ABUS, or used internally by other blocks. These settings can be programmed in the VDAC_OUTCTRL register after VDAC is enabled but ideally before channels are enabled. The possible settings are as follows:

- VDAC_OUTCTRL.MAINOUTENCHx - Set this bit to route VDAC channel analog main output to the dedicated pin (CHx_MAIN_OUT). This is the preferred option for any DAC output.
- VDAC_OUTCTRL.AUXOUTENCHx - Set this bit to route VDAC channel analog auxiliary output to internal blocks such as IADC and ACMP, or to the ABUS interconnect matrix, where it can be routed to any port I/O supporting ABUS. Note that the ABUS multiplexer adds significant impedance, and this option may not be suitable for certain loads or dynamic conditions.
- VDAC_OUTCTRL.SHORTCHx - Set this bit when using the high-power buffer option with the auxiliary outputs. This will short the MAIN and AUX outputs together. The MAIN output is still driven in this configuration.
- VDAC_OUTCTRL.ABUSPORTSELCHx - Select a particular ABUS port for the channel. Choose values from PORTA, PORTB, PORTC, PORTD
- VDAC_OUTCTRL.ABUSPINSELCHx - Select a particular ABUS pin for the channel. Program the pin in conjunction with the port to route the analog output to a particular GPIO.

Note: Once the channel is enabled, the output control settings in VDAC and the ABUS configuration in both VDAC and GPIO should not be changed until the channel is disabled. Changing these settings while enabled may cause spurious voltage generation on random GPIO.

ABUS Allocation Rules to VDAC in GPIO

- AEVEN0/BEVEN0/CDEVEN0 can only be allocated to CH0
- AEVEN1/BEVEN1/CDEVEN1 can only be allocated to CH1
- AODD0/BODD0/CDODD0 can only be allocated to CH0
- AODD1/BODD1/CDODD1 can only be allocated to CH1
- The port and pin requested by ABUSPORTSELCHx/ABUSPINSELCHx must match a bus allocated to VDAC CHx.

  For example, if AEVEN0 is the only bus allocated to CH0, the selected port must be port A, and the selected pin must be even.

More details on ABUS allocation can be found in GPIO chapter

28.4 VDAC Register Map

The offset register address is relative to the registers base address.

Offset	Name	Type	Description
0x000	VDAC_IPVERSION	R	IPVERSION
0x004	VDAC_EN	RW ENABLE	Module Enable
0x008	VDAC_SWRST	RW SWRST	Software Reset Register
0x00C	VDAC_CFG	RW CONFIG	Config Register
0x010	VDAC_STATUS	RH	Status Register
0x014	VDAC_CH0CFG	RW CONFIG	Channel 0 Config Register
0x018	VDAC_CH1CFG	RW CONFIG	Channel 1 Config Register
0x01C	VDAC_CMD	W SYNC	Command Register
0x020	VDAC_IF	RWH INTFLAG	Interrupt Flag Register
0x024	VDAC_IEN	RW	Interrupt Enable Register
0x028	VDAC_CH0F	W	Channel 0 Data Write Fifo
0x02C	VDAC_CH1F	W	Channel 1 Data Write Fifo
0x030	VDAC_OUTCTRL	RW SYNC	DAC Output Control
0x034	VDAC_OUTTIMERCFG	RW CONFIG	DAC Out Timer Config Register
0x1000	VDAC_IPVERSION_SET	R	IPVERSION
0x1004	VDAC_EN_SET	RW ENABLE	Module Enable
0x1008	VDAC_SWRST_SET	RW SWRST	Software Reset Register
0x100C	VDAC_CFG_SET	RW CONFIG	Config Register
0x1010	VDAC_STATUS_SET	RH	Status Register
0x1014	VDAC_CH0CFG_SET	RW CONFIG	Channel 0 Config Register
0x1018	VDAC_CH1CFG_SET	RW CONFIG	Channel 1 Config Register
0x101C	VDAC_CMD_SET	W SYNC	Command Register
0x1020	VDAC_IF_SET	RWH INTFLAG	Interrupt Flag Register
0x1024	VDAC_IEN_SET	RW	Interrupt Enable Register
0x1028	VDAC_CH0F_SET	W	Channel 0 Data Write Fifo
0x102C	VDAC_CH1F_SET	W	Channel 1 Data Write Fifo
0x1030	VDAC_OUTCTRL_SET	RW SYNC	DAC Output Control
0x1034	VDAC_OUTTIMERCFG_SET	RW CONFIG	DAC Out Timer Config Register
0x2000	VDAC_IPVERSION_CLR	R	IPVERSION
0x2004	VDAC_EN_CLR	RW ENABLE	Module Enable
0x2008	VDAC_SWRST_CLR	RW SWRST	Software Reset Register
0x200C	VDAC_CFG_CLR	RW CONFIG	Config Register
0x2010	VDAC_STATUS_CLR	RH	Status Register
0x2014	VDAC_CH0CFG_CLR	RW CONFIG	Channel 0 Config Register
0x2018	VDAC_CH1CFG_CLR	RW CONFIG	Channel 1 Config Register

Offset	Name	Type	Description
0x201C	VDAC_CMD_CLR	W SYNC	Command Register
0x2020	VDAC_IF_CLR	RWH INTFLAG	Interrupt Flag Register
0x2024	VDAC_IEN_CLR	RW	Interrupt Enable Register
0x2028	VDAC_CH0F_CLR	W	Channel 0 Data Write Fifo
0x202C	VDAC_CH1F_CLR	W	Channel 1 Data Write Fifo
0x2030	VDAC_OUTCTRL_CLR	RW SYNC	DAC Output Control
0x2034	VDAC_OUTTIMERCFG_CLR	RW CONFIG	DAC Out Timer Config Register
0x3000	VDAC_IPVERSION_TGL	R	IPVERSION
0x3004	VDAC_EN_TGL	RW ENABLE	Module Enable
0x3008	VDAC_SWRST_TGL	RW SWRST	Software Reset Register
0x300C	VDAC_CFG_TGL	RW CONFIG	Config Register
0x3010	VDAC_STATUS_TGL	RH	Status Register
0x3014	VDAC_CH0CFG_TGL	RW CONFIG	Channel 0 Config Register
0x3018	VDAC_CH1CFG_TGL	RW CONFIG	Channel 1 Config Register
0x301C	VDAC_CMD_TGL	W SYNC	Command Register
0x3020	VDAC_IF_TGL	RWH INTFLAG	Interrupt Flag Register
0x3024	VDAC_IEN_TGL	RW	Interrupt Enable Register
0x3028	VDAC_CH0F_TGL	W	Channel 0 Data Write Fifo
0x302C	VDAC_CH1F_TGL	W	Channel 1 Data Write Fifo
0x3030	VDAC_OUTCTRL_TGL	RW SYNC	DAC Output Control
0x3034	VDAC_OUTTIMERCFG_TGL	RW CONFIG	DAC Out Timer Config Register

28.5 VDAC Register Description

28.5.1 VDAC_IPVERSION - IPVERSION

Offset	Bit Position																															
0x000	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset	0x2																															
Access	R																															
Name	IPVERSION																															

Bit	Name	Reset	Access	Description
31:0	IPVERSION	0x2	R	**IPVERSION**
				The read only IPVERSION field gives the version for this module. There may be minor software changes required for modules with different values of IPVERSION.

28.5.2 VDAC_EN - Module Enable

Offset	Bit Position																															
0x004	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset																															0x0	0x0
Access																															R	RW
Name																															DISABLING	EN

Bit	Name	Reset	Access	Description
31:2	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
1	DISABLING	0x0	R	**Disablement busy status**
	When EN is cleared, DISABLING is set immediately, and cleared when disablement finishes. Disablement resets peripheral cores and not APB registers except hardware updated registers such as INTFLAGs and FIFOs.			
0	EN	0x0	RW	**VDAC Module Enable**
	Enable the VDAC module. When EN is cleared(disablement), it halts module operation immediately, and initialize the core domain such that when the is re-enabled, it starts cleanly.			

Value	Mode	Description
0	DISABLE	Disable
1	ENABLE	Enable

28.5.3 VDAC_SWRST - Software Reset Register

Offset	Bit Position																															
0x008	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset																															0x0	0x0
Access																															R	W
Name																															RESETTING	SWRST

Bit	Name	Reset	Access	Description
31:2	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
1	RESETTING	0x0	R	**Software reset busy status**
	When SWRST command is issued, resetting logic sets RESETTING status immediately, and later it is cleared when reset process finishes.			
0	SWRST	0x0	W	**Software reset command**
	A software reset command field resets the module back to the initial condition, similar to a power on reset condition			

28.5.4 VDAC_CFG - Config Register

Offset	Bit Position																															
0x00C	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset		0x2			0x0	0x0	0x0	0x0		0x0				0x0					0x0								0x0		0x0	0x0	0x0	0x0
Access		RW			RW	RW	RW	RW		RW				RW					RW								RW		RW	RW	RW	RW
Name		WARMUPTIME			DBGHALT	ONDEMANDCLK	DMAWU	BIASKEEPWARM		REFRESHPERIOD				TIMEROVRFLOWPERIOD					PRESC								REFRSEL		CH0PRESCRST	SINERESET	SINEMODE	DIFF

Bit	Name	Reset	Access	Description
31	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
30:28	WARMUPTIME	0x2	RW	**DAC Warmup Time**
	Number of prescaled CLK_DAC +1 cycles for the VDAC to Warmup. Default is (2+1) prescaled CLK_DAC cycles.			
27	DBGHALT	0x0	RW	**Debug Halt**
	VDAC behavior when halted by debugger			

Value	Mode	Description
0	NORMAL	Continue operation as normal during debug mode
1	HALT	Complete the current conversion and then halt during debug mode

Bit	Name	Reset	Access	Description
26	ONDEMANDCLK	0x0	RW	**Always allow clk_dac**
	Setting this bit to 1 always allows CLK_DAC from CMU			
25	DMAWU	0x0	RW	**VDAC DMA Wakeup**
	Set to enable wakeup from EM2 to EM1 for DMA to fill CHx FIFO Data			
24	BIASKEEPWARM	0x0	RW	**Bias Keepwarm Mode Enable**
	Set this bit to keep the Bias on for Analog portion of DAC in between conversions. Primary purpose, is to reduce kick-back to the reference shared with IADC. Relevant only for Sample-off Mode			
23	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
22:20	REFRESHPERIOD	0x0	RW	**Refresh Timer Overflow Period**
	Select refresh counter period. A channel x will be refreshed with the period set in REFRESHPERIOD if the channel in VDACn_CHxCFG has its REFRESHSOURCE set to REFRESHTIMER			

Value	Mode	Description
0	CYCLES2	All channels with enabled refresh are refreshed every 2 CLK_REFRESH cycles

Bit	Name	Reset	Access	Description
	1	CYCLES4		All channels with enabled refresh are refreshed every 4 CLK_REFRESH cycles
	2	CYCLES8		All channels with enabled refresh are refreshed every 8 CLK_REFRESH cycles
	3	CYCLES16		All channels with enabled refresh are refreshed every 16 CLK_REFRESH cycles
	4	CYCLES32		All channels with enabled refresh are refreshed every 32 CLK_REFRESH cycles
	5	CYCLES64		All channels with enabled refresh are refreshed every 64 CLK_REFRESH cycles
	6	CYCLES128		All channels with enabled refresh are refreshed every 128 CLK_REFRESH cycles
	7	CYCLES256		All channels with enabled refresh are refreshed every 256 CLK_REFRESH cycles
19	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
18:16	TIMEROVRFLOWPERIOD	0x0	RW	**Internal Timer Overflow Period**
	Select internal timer overflow period. A channel x will be provided with a conversion trigger after the period set in TIMEROVRFLOWPERIOD if the channel in VDACn_CHxCFG has its TRIGMODE set to INTERNALTIMER			
	Value	Mode		Description
	0	CYCLES2		The Timer overflows every 2 Prescaled CLK_DAC cycles
	1	CYCLES4		The Timer overflows every 4 Prescaled CLK_DAC cycles
	2	CYCLES8		The Timer overflows every 8 Prescaled CLK_DAC cycles
	3	CYCLES16		The Timer overflows every 16 Prescaled CLK_DAC cycles
	4	CYCLES32		The Timer overflows every 32 Prescaled CLK_DAC cycles
	5	CYCLES64		The Timer overflows every 64 Prescaled CLK_DAC cycles
15:14	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
13:7	PRESC	0x0	RW	**Prescaler Setting for DAC clock**
	Selected DAC clock source is prescaled by PRESC+1 to generate prescaled CLK_DAC with 50% duty cycle			
6	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
5:4	REFRSEL	0x0	RW	**Reference Selection**
	Select reference for analog portion of DAC			
	Value	Mode		Description
	0	V125		Internal 1.25 V bandgap reference
	1	V25		Internal 2.5 V bandgap reference
	2	VDD		AVDD reference
	3	EXT		External pin reference

Bit	Name	Reset	Access	Description
3	CH0PRESCRST	0x0	RW	**Channel 0 Start Reset Prescaler**
	Select if prescaler (determining prescaled CLK_DAC rate) is reset on channel 0 start.			
	Value	Mode		Description
	0	NORESETPRESC		Prescaler not reset on channel 0 start
	1	RESETPRESC		Prescaler reset on channel 0 start
2	SINERESET	0x0	RW	**Sine Wave Reset When inactive**
	In case SINERESET is 0, SINEMODESTOP will stop progress of the sine wave at the sample currently being output in sinemode. When Set to 1,the sine output will reset to 0 degrees when SINEMODESTOP			
1	SINEMODE	0x0	RW	**Sine Mode**
	Enable/disable sine mode.			
	Value	Mode		Description
	0	DISSINEMODE		Sine mode disabled. Sine reset to 0 degrees
	1	ENSINEMODE		Sine mode enabled
0	DIFF	0x0	RW	**Differential Mode**
	Select single ended or differential mode.			
	Value	Mode		Description
	0	SINGLEENDED		Single ended output
	1	DIFFERENTIAL		Differential output

28.5.5 VDAC_STATUS - Status Register

Offset	Bit Position																															
0x010	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset	0x0	0x0	0x0	0x0	0x0	0x0			0x0	0x0		0x0	0x0		0x0			0x0					0x0	0x0			0x0	0x0			0x0	0x0
Access	R	R	R	R	R	R			R	R		R	R		R			R					R	R			R	R			R	R
Name	SYNCBUSY	ABUSALLOCERR	SINEACTIVE	ABUSINPUTCONFLICT	CH1FIFOFLBUSY	CH0FIFOFLBUSY			CH1FIFOEMPTY	CH0FIFOEMPTY		CH1CURRENTSTATE	CH0CURRENTSTATE		CH1FIFOCNT			CH0FIFOCNT					CH1FIFOFULL	CH0FIFOFULL			CH1WARM	CH0WARM			CH1ENS	CH0ENS

Bit	Name	Reset	Access	Description
31	SYNCBUSY	0x0	R	**Sync Busy Combined**
	Indicates synchronization ongoing			
30	ABUSALLOCERR	0x0	R	**ABUS Allocation Error Status**
	1 if ABUS requested is not allocated			
29	SINEACTIVE	0x0	R	**Sine Wave Output Status on Channel**
	1 indicates that sine wave output is generated on CH0 and/or CH1(in diff mode)			
28	ABUSINPUTCONFLICT	0x0	R	**ABUS Input Conflict Status**
	1 if both CH0 and CH1 request the same ABUS			
27	CH1FIFOFLBUSY	0x0	R	**CH1 FIFO Flush Sync Busy**
	Indicates CH1 FIFO Flush Sync Busy			
26	CH0FIFOFLBUSY	0x0	R	**CH0 FIFO Flush Sync Busy**
	Indicates CH0 FIFO Flush Sync Busy			
25:24	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
23	CH1FIFOEMPTY	0x0	R	**Channel 1 FIFO Empty Status**
	1 if FIFO for Channel 1 is empty			
22	CH0FIFOEMPTY	0x0	R	**Channel 0 FIFO Empty Status**
	1 if FIFO for Channel 0 is empty			
21	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
20	CH1CURRENTSTATE	0x0	R	**Channel 1 Current Status**
	0 when CH1 is in IDLE State. Set when CH1 is converting data			
19	CH0CURRENTSTATE	0x0	R	**Channel 0 Current Status**
	0 when CH0 is in IDLE State. Set when CH0 is converting data			
18	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
17:15	CH1FIFOCNT	0x0	R	**Channel 1 FIFO Valid Count**

Bit	Name	Reset	Access	Description
	Number of Valid Entries in Channel 1. This FIFO entries Count is generated in Read Domain hence expect Synchronization Delay. Need to be used only when VDAC_CFG.ONDEMANDCLK is set to 1.			
14:12	CH0FIFOCNT	0x0	R	**Channel 0 FIFO Valid Count**
	Number of Valid Entries in Channel 0. This FIFO entries Count is generated in Read Domain hence expect Synchronization Delay. Need to be used only when VDAC_CFG.ONDEMANDCLK is set to 1.			
11:10	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
9	CH1FIFOFULL	0x0	R	**Channel 1 FIFO Full Status**
	1 if FIFO for Channel 1 is full			
8	CH0FIFOFULL	0x0	R	**Channel 0 FIFO Full Status**
	1 if FIFO for Channel 0 is full			
7:6	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
5	CH1WARM	0x0	R	**Channel 1 Warmed Status**
	This bit is set when channel 1 has warmed up			
4	CH0WARM	0x0	R	**Channel 0 Warmed Status**
	This bit is set when channel 0 has warmed up			
3:2	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
1	CH1ENS	0x0	R	**Channel 1 Enabled Status**
	This bit is set when channel 1 is enabled.			
0	CH0ENS	0x0	R	**Channel 0 Enabled Status**
	This bit is set when channel 0 is enabled.			

28.5.6 VDAC_CH0CFG - Channel 0 Config Register

Offset	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x014																																
Reset																0x0		0x0		0x0			0x0			0x0				0x0		0x0
Access																RW		RW		RW			RW			RW				RW		RW
Name																KEEPWARM		HIGHCAPLOADEN		FIFODVL			REFRESHSOURCE			TRIGMODE				POWERMODE		CONVMODE

Bit	Name	Reset	Access	Description
31:17	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
16	KEEPWARM	0x0	RW	**Channel 0 Keepwarm Mode Enable**
	Set this bit to keep the Channel 0 on in between conversion in sample-off mode. Primary purpose of this is to reduce kickback between Channel 0 and Channel 1 and to reduce startup time.			
15	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
14	HIGHCAPLOADEN	0x0	RW	**Channel 0 High Cap Load Mode Enable**
	Enables High Capacitance Load Mode for Channel 0. Should be enabled with VDAC_CH0CFG.POWERMODE=HIGHPOWER			
13	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
12:11	FIFODVL	0x0	RW	**Channel 0 FIFO Low Watermark**
	Set Channel 0 FIFO Low Threshold Data Valid Level (DVL)			
10	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
9:8	REFRESHSOURCE	0x0	RW	**Channel 0 Refresh Source**
	Select Channel 0 Refresh Trigger			

Value	Mode	Description
0	NONE	No Refresh Source Selected for Channel 0.
1	REFRESHTIMER	Channel 0 Refresh triggered by Refresh Timer Overflow
2	SYNCPRS	Channel 0 Refresh triggered by Sync PRS. PRS Trigger should have the same clock group as VDAC.
3	ASYNCPRS	Channel 0 Refresh triggered by Async PRS

Bit	Name	Reset	Access	Description
7	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
6:4	TRIGMODE	0x0	RW	**Channel 0 Trigger Mode**
	Select Channel 0 Conversion Trigger			

Bit	Name	Reset	Access	Description
	Value	Mode		Description
	0	NONE		No Conversion Trigger Source Selected for Channel 0
	1	SW		Channel 0 is triggered by Channel 0 FIFO (CH0F) write
	2	SYNCPRS		Channel 0 is triggered by Sync PRS input. PRS Trigger should have the same clock group as VDAC.
	4	INTERNALTIMER		Channel 0 is triggered by Internal Timer Overflow
	5	ASYNCPRS		Channel 0 is triggered by Async PRS input
3	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
2	POWERMODE	0x0	RW	**Channel 0 Power Mode**
	Enable Power Mode for Channel 0			
	Value	Mode		Description
	0	HIGHPOWER		Default is High Power Mode
	1	LOWPOWER		Set this bit for Low Power Mode
1	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
0	CONVMODE	0x0	RW	**Channel 0 Conversion Mode**
	Configure conversion mode			
	Value	Mode		Description
	0	CONTINUOUS		DAC channel 0 is set in continuous mode
	1	SAMPLEOFF		DAC channel 0 is set in sample/shut off mode

28.5.7 VDAC_CH1CFG - Channel 1 Config Register

Offset	Bit Position																														
0x018	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset																0x0		0x0		0x0			0x0			0x0				0x0		0x0
Access																RW		RW		RW			RW			RW				RW		RW
Name																KEEPWARM		HIGHCAPLOADEN		FIFODVL			REFRESHSOURCE			TRIGMODE				POWERMODE		CONVMODE

Bit	Name	Reset	Access	Description
31:17	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
16	KEEPWARM	0x0	RW	**Channel 1 Keepwarm Mode Enable**
	Set this bit to keep the Channel 1 on in between conversion in sample-off mode. Primary purpose of this is to reduce kickback between Channel 0 and Channel 1 and to reduce startup time.			
15	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
14	HIGHCAPLOADEN	0x0	RW	**Channel 1 High Cap Load Mode Enable**
	Enables High Capacitance Load Mode for Channel 1. Should be enabled with VDAC_CH1CFG.POWERMODE=HIGHPOWER.			
13	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
12:11	FIFODVL	0x0	RW	**Channel 1 FIFO Low Watermark**
	Set Channel 1 Low threshold Data Valid Level (DVL)			
10	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
9:8	REFRESHSOURCE	0x0	RW	**Channel 1 Refresh Source**
	Select Channel 1 Refresh Trigger			

Value	Mode	Description
0	NONE	No Refresh Source Selected
1	REFRESHTIMER	CH1 Refresh Triggered by Refresh Timer Overflow
2	SYNCPRS	CH1 Refresh Triggered by Sync PRS. PRS Trigger should have the same clock group as VDAC.
3	ASYNCPRS	CH1 Refresh Triggered by Async PRS

Bit	Name	Reset	Access	Description
7	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
6:4	TRIGMODE	0x0	RW	**Channel 1 Trigger Mode**
	Select Channel 1 Conversion Trigger			

Bit	Name	Reset	Access	Description
	Value	Mode		Description
	0	NONE		No Conversion Trigger Source Selected for Channel 1
	1	SW		Channel 1 is triggered by Channel 1 FIFO (CH1F) write
	2	SYNCPRS		Channel 1 is triggered by Sync PRS input.PRS Trigger should have the same clock group as VDAC.
	4	INTERNALTIMER		Channel 1 is triggered by Internal Timer Overflow
	5	ASYNCPRS		Channel 1 is triggered by Async PRS input
3	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
2	POWERMODE	0x0	RW	**Channel 1 Power Mode**
	Enable Low Power Mode for Channel 1			
	Value	Mode		Description
	0	HIGHPOWER		Default is High Power Mode
	1	LOWPOWER		Set this bit for Low Power Mode
1	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
0	CONVMODE	0x0	RW	**Channel 1 Conversion Mode**
	Configure conversion mode for Channel 1			
	Value	Mode		Description
	0	CONTINUOUS		DAC channel 1 is set in continuous mode
	1	SAMPLEOFF		DAC channel 1 is set in sample/shut off mode

28.5.8 VDAC_CMD - Command Register

Offset	Bit Position																															
0x01C	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset																					0x0	0x0	0x0	0x0			0x0	0x0			0x0	0x0
Access																					W(nB)	W(nB)	W(nB)	W(nB)			W(nB)	W(nB)			W(nB)	W(nB)
Name																					SINEMODESTOP	SINEMODESTART	CH1FIFOFLUSH	CH0FIFOFLUSH			CH1DIS	CH1EN			CH0DIS	CH0EN

Bit	Name	Reset	Access	Description
31:12	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
11	SINEMODESTOP	0x0	W(nB)	**Stop Sine Wave Generation**
	Stop Sine Wave Generation			
10	SINEMODESTART	0x0	W(nB)	**Start Sine Wave Generation**
	Start Sine Wave Generation			
9	CH1FIFOFLUSH	0x0	W(nB)	**CH1 WFIFO Flush**
	Flush Channel 1 WFIFO			
8	CH0FIFOFLUSH	0x0	W(nB)	**CH0 WFIFO Flush**
	Flush Channel 0 WFIFO			
7:6	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
5	CH1DIS	0x0	W(nB)	**DAC Channel 1 Disable**
	Disables DAC Channel 1			
4	CH1EN	0x0	W(nB)	**DAC Channel 1 Enable**
	Enables DAC Channel 1			
3:2	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
1	CH0DIS	0x0	W(nB)	**DAC Channel 0 Disable**
	Disables DAC Channel 0			
0	CH0EN	0x0	W(nB)	**DAC Channel 0 Enable**
	Enables DAC Channel 0			

28.5.9 VDAC_IF - Interrupt Flag Register

Offset	Bit Position																															
0x020	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset						0x0					0x0	0x0		0x0									0x0	0x0			0x0	0x0			0x0	0x0
Access						RW					RW	RW		RW									RW	RW			RW	RW			RW	RW
Name						ABUSINPUTCONFLICT					CH1DVL	CH0DVL		ABUSALLOCERR									CH1UF	CH0UF			CH1OF	CH0OF			CH1CD	CH0CD

Bit	Name	Reset	Access	Description
31:27	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
26	ABUSINPUTCONFLICT	0x0	RW	**ABUS Input Conflict Error Flag**
	Set if both CH0 and CH1 request same ABUS. Should only be enabled when using ABUS for VDAC Output.			
25:22	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
21	CH1DVL	0x0	RW	**CH1 Data Valid Level Interrupt Flag**
	Set when Channel 1 FIFO Count reaches Data Valid Level. Also used as Wakeup IRQ			
20	CH0DVL	0x0	RW	**CH0 Data Valid Level Interrupt Flag**
	Set when Channel 0 FIFO Count reaches Data Valid Level. Also used as Wakeup IRQ			
19	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
18	ABUSALLOCERR	0x0	RW	**ABUS Port Allocation Error Flag**
	Set if APORT requested is not allocated. Should only be enabled when using ABUS for VDAC Output.			
17:10	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
9	CH1UF	0x0	RW	**CH1 Data Underflow Interrupt Flag**
	Indicates channel 1 data underflow.			
8	CH0UF	0x0	RW	**CH0 Data Underflow Interrupt Flag**
	Indicates channel 0 data underflow.			
7:6	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
5	CH1OF	0x0	RW	**CH1 Data Overflow Interrupt Flag**
	Indicates channel 1 data overflow.			
4	CH0OF	0x0	RW	**CH0 Data Overflow Interrupt Flag**
	Indicates channel 0 data overflow.			
3:2	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		

Bit	Name	Reset	Access	Description
1	CH1CD	0x0	RW	**CH1 Conversion Done Interrupt Flag**
	Indicates channel 1 conversion complete.			
0	CH0CD	0x0	RW	**CH0 Conversion Done Interrupt Flag**
	Indicates channel 0 conversion complete.			

28.5.10 VDAC_IEN - Interrupt Enable Register

Offset	Bit Position																															
0x024	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset						0x0					0x0	0x0		0x0									0x0	0x0			0x0	0x0			0x0	0x0
Access						RW					RW	RW		RW									RW	RW			RW	RW			RW	RW
Name						ABUSINPUTCONFLICT					CH1DVL	CH0DVL		ABUSALLOCERR									CH1UF	CH0UF			CH1OF	CH0OF			CH1CD	CH0CD

Bit	Name	Reset	Access	Description
31:27	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
26	ABUSINPUTCONFLICT	0x0	RW	**ABUS Input Conflict Interrupt Flag**
	Set to enable the ABUSINPUTCONFLICTIF Interrupt			
25:22	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
21	CH1DVL	0x0	RW	**CH1 Data Valid Level Interrupt Flag**
	Set to enable the CH1DVLIF Interrupt			
20	CH0DVL	0x0	RW	**CH0 Data Valid Level Interrupt Flag**
	Set to enable the CH0DVLIF Interrupt			
19	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
18	ABUSALLOCERR	0x0	RW	**ABUS Allocation Error Interrupt Flag**
	Set to enable the ABUSALLOCERRIF Interrupt			
17:10	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
9	CH1UF	0x0	RW	**CH1 Data Underflow Interrupt Flag**
	Set to enable the CH1UFIF Interrupt			
8	CH0UF	0x0	RW	**CH0 Data Underflow Interrupt Flag**
	Set to enable the CH0UFIF Interrupt			
7:6	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
5	CH1OF	0x0	RW	**CH1 Data Overflow Interrupt Flag**
	Set to enable the CH1OFIF Interrupt			
4	CH0OF	0x0	RW	**CH0 Data Overflow Interrupt Flag**
	Set to enable the CH0OFIF Interrupt			
3:2	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		

Bit	Name	Reset	Access	Description
1	CH1CD	0x0	RW	**CH1 Conversion Done Interrupt Flag**
	Set to enable the CH1CDIF Interrupt			
0	CH0CD	0x0	RW	**CH0 Conversion Done Interrupt Flag**
	Set to enable the CH0CDIF Interrupt			

28.5.11 VDAC_CH0F - Channel 0 Data Write Fifo

Offset	Bit Position	
0x028	31 … 12	11 … 0
Reset		0x0
Access		W
Name		DATA

Bit	Name	Reset	Access	Description
31:12	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
11:0	DATA	0x0	W	**Channel 0 Data**
	This register writes the value which will be converted by DAC channel 0 in a FIFO.			

28.5.12 VDAC_CH1F - Channel 1 Data Write Fifo

Offset	Bit Position	
0x02C	31 … 12	11 … 0
Reset		0x0
Access		W
Name		DATA

Bit	Name	Reset	Access	Description
31:12	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
11:0	DATA	0x0	W	**Channel 1 Data**
	This register writes the value which will be converted by DAC channel 1 into a FIFO			

28.5.13 VDAC_OUTCTRL - DAC Output Control

Offset	Bit Position																															
0x030	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset		0x0						0x0				0x0						0x0					0x0	0x0			0x0	0x0			0x0	0x0
Access		RW						RW				RW						RW					RW	RW			RW	RW			RW	RW
Name		ABUSPINSELCH1						ABUSPORTSELCH1				ABUSPINSELCH0						ABUSPORTSELCH0					SHORTCH1	SHORTCH0			AUXOUTENCH1	AUXOUTENCH0			MAINOUTENCH1	MAINOUTENCH0

Bit	Name	Reset	Access	Description
31	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
30:25	ABUSPINSELCH1	0x0	RW	**CH1 ABUS Pin Select**
	Set this to Select a particular ABUS Pin for Channel 1			
24:22	ABUSPORTSELCH1	0x0	RW	**CH1 ABUS Port Select**
	Set this to Select a particular ABUS Port for Channel 1			

Value	Mode	Description
0	NONE	No GPIO Selected for CH1 ABUS Output
1	PORTA	Port A Selected
2	PORTB	Port B Selected
3	PORTC	Port C Selected
4	PORTD	Port D Selected

Bit	Name	Reset	Access	Description
21	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
20:15	ABUSPINSELCH0	0x0	RW	**CH0 ABUS Pin Select**
	Set this to Select a particular ABUS Pin for Channel 0			
14:12	ABUSPORTSELCH0	0x0	RW	**CH0 ABUS Port Select**
	Set this to Select a particular ABUS Port for Channel 0			

Value	Mode	Description
0	NONE	No GPIO Selected for CH0 ABUS Output
1	PORTA	Port A Selected
2	PORTB	Port B Selected
3	PORTC	Port C Selected
4	PORTD	Port D Selected

Bit	Name	Reset	Access	Description
11:10	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		

Bit	Name	Reset	Access	Description
9	SHORTCH1	0x0	RW	**CH0 Main and Alternative Output Short**
	Set this to short circuit Main and alternative Output of Channel 0. Usefull to connect Main and Alternative outputs together when DAC is enabled but not warmed up yet.			
8	SHORTCH0	0x0	RW	**CH1 Main and Alternative Output Short**
	Set this to short circuit Main and alternative Output of Channel 1. Usefull to connect Main and Alternative outputs together when DAC is enabled but not warmed up yet.			
7:6	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
5	AUXOUTENCH1	0x0	RW	**CH1 Alternative Output Enable**
	Set this to enable alternative output of Channel 1			
4	AUXOUTENCH0	0x0	RW	**CH0 Alternative Output Enable**
	Set this to enable alternative output of Channel 0			
3:2	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
1	MAINOUTENCH1	0x0	RW	**CH1 Main Output Enable**
	Set this to enable main output of Channel 1			
0	MAINOUTENCH0	0x0	RW	**CH0 Main Output Enable**
	Set this to enable main output of Channel 0			

28.5.14 VDAC_OUTTIMERCFG - DAC Out Timer Config Register

Offset	Bit Position																															
0x034	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset												0x0															0x0					
Access												RW															RW					
Name												CH1OUTHOLDTIME															CH0OUTHOLDTIME					

Bit	Name	Reset	Access	Description
31:25	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
24:15	CH1OUTHOLDTIME	0x0	RW	**CH1 Output Hold Time**
	Number of prescaled CLK_DAC cycles to drive the output of VDAC for Channel 1			
14:10	*Reserved*	*To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions*		
9:0	CH0OUTHOLDTIME	0x0	RW	**CH0 Output Hold Time**
	Number of prescaled CLK_DAC cycles to drive the output of VDAC for Channel 0			